UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

Class 011 · Lagrange's method of undetermined multipliers · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

Contents

1. Figure F1.1 · Lagrange's working procedure
2. Figure F1.2 · Choosing f and ϕ
3. Elimination (class 010) and Lagrange's method

1 Lagrange's method: the auxiliary function and the procedure

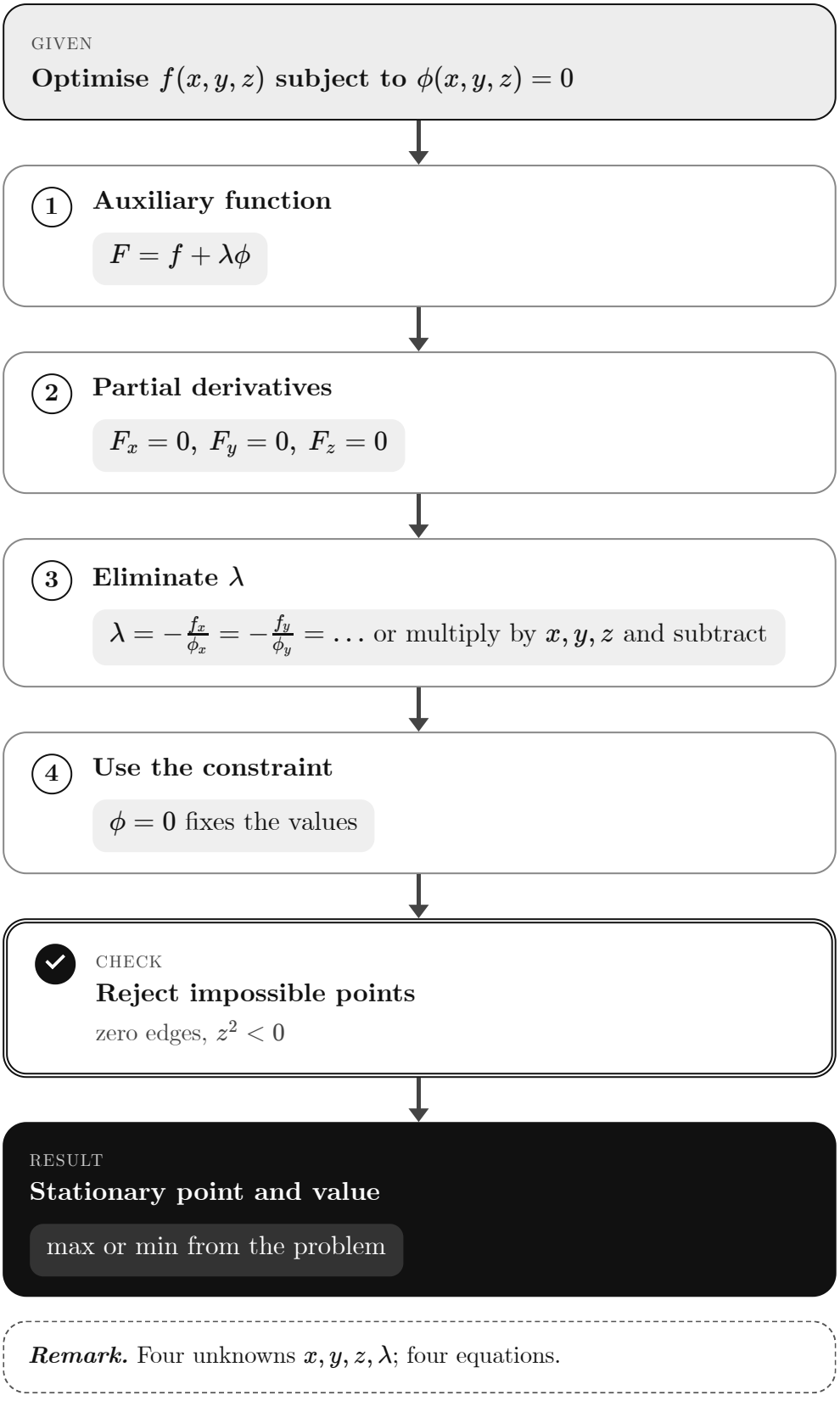


Figure F1.1. Lagrange's working procedure ESA ★★★

S011·3 · T011a 00:01:18

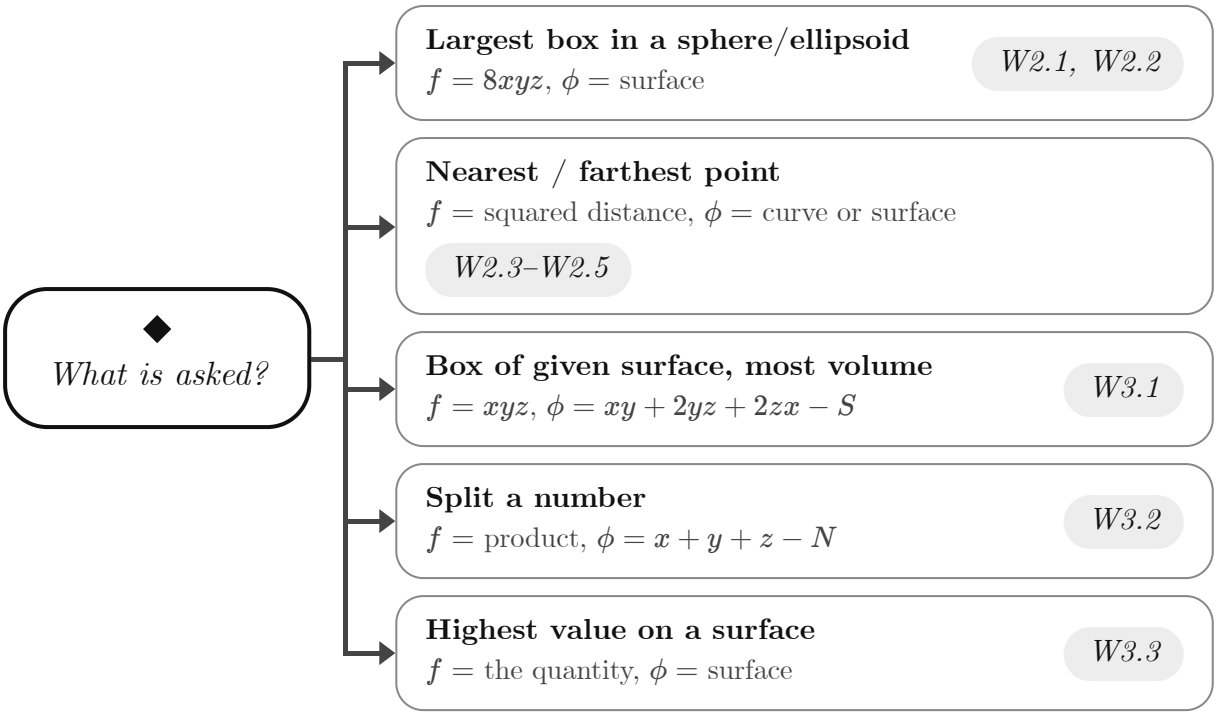


Figure F1.2. Choosing f and ϕ ESA ★★★

T011a 00:09:15 · T011b 00:01:46 · S011·10 · S011·12

Elimination (class 010) and Lagrange's method

	Elimination	Lagrange's multipliers
Constraint used	Solved for one variable and substituted	Added as $\lambda\phi$
Equations	$f_x = f_y = 0$ in two variables	$F_x = F_y = F_z = 0$, $\phi = 0$
Nature of the point	$rt - s^2$ and r decide	Not decided; from the problem
More variables / constraints	Hard	Extends directly
Temperature $400xyz^2$	50 (class 010)	50 (W3.3)

S011·4 · T011a 00:01:36 · T011a 00:04:23